

09

1

Describe the mass flow hypothesis for the mechanism of translocation in plants.

[4 marks]

[Extra space]

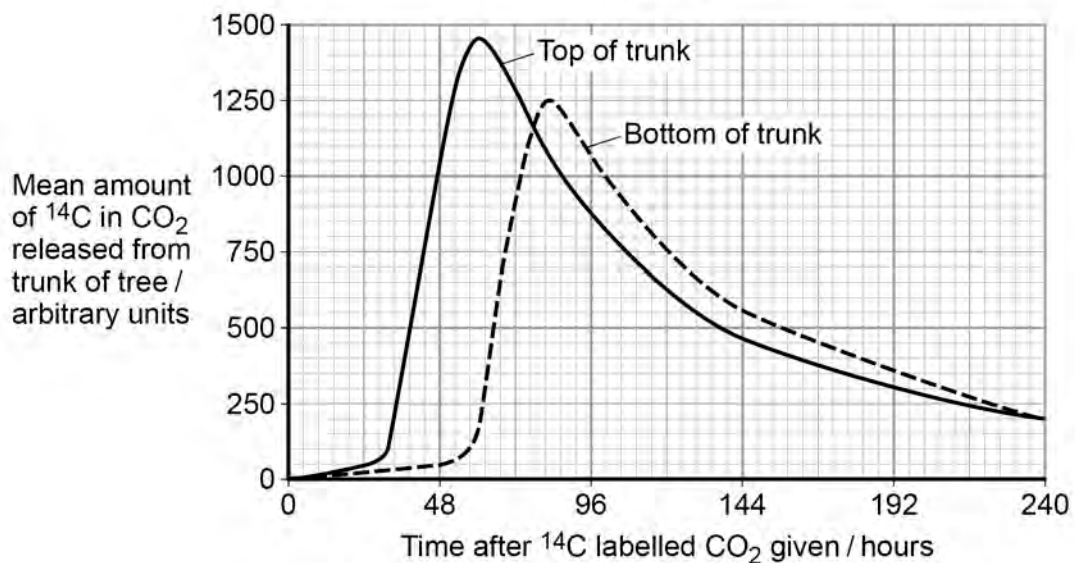
Scientists measured translocation in the phloem of trees. They used carbon dioxide labelled with radioactive ^{14}C .

They put a large, clear plastic bag over the leaves and branches of each tree and added $^{14}\text{CO}_2$. The main trunk of the tree was not in the plastic bag.

At regular intervals after adding the $^{14}\text{CO}_2$ to the bag, the scientists measured the amount of $^{14}\text{CO}_2$ released from the top and bottom of the main trunk of the tree. On the surface of the trunk of these trees, there are pores for gas exchange.

Figure 7 shows the scientists' results.

Figure 7



0 9 . **2** Name the process that produced the $^{14}\text{CO}_2$ released from the trunk.

[1 mark]

0 9 . **3** How long did it take the ^{14}C label to get from the top of the trunk to the bottom of the trunk? Explain how you reached your answer.

[2 marks]

0 9 . **4** What other information is required in order to calculate the mean rate of movement of the ^{14}C down the trunk?

[1 mark]

Turn over for the next question
